M³ provisional FeO calculation

Reproducible diagnostic run and next validation gates | 18 September 2026

Can this image tell us whether a region contains iron

Not yet. This figure displays measured optical behavior, not an iron detection or an abundance estimate. The useful question is whether the spectra contain reliable evidence of Fe-bearing material and whether a tested calibration can estimate its bulk FeO-equivalent concentration. No part of this image can presently be labeled iron-rich or iron-poor on the basis of this analysis alone. [1, 4, 5]

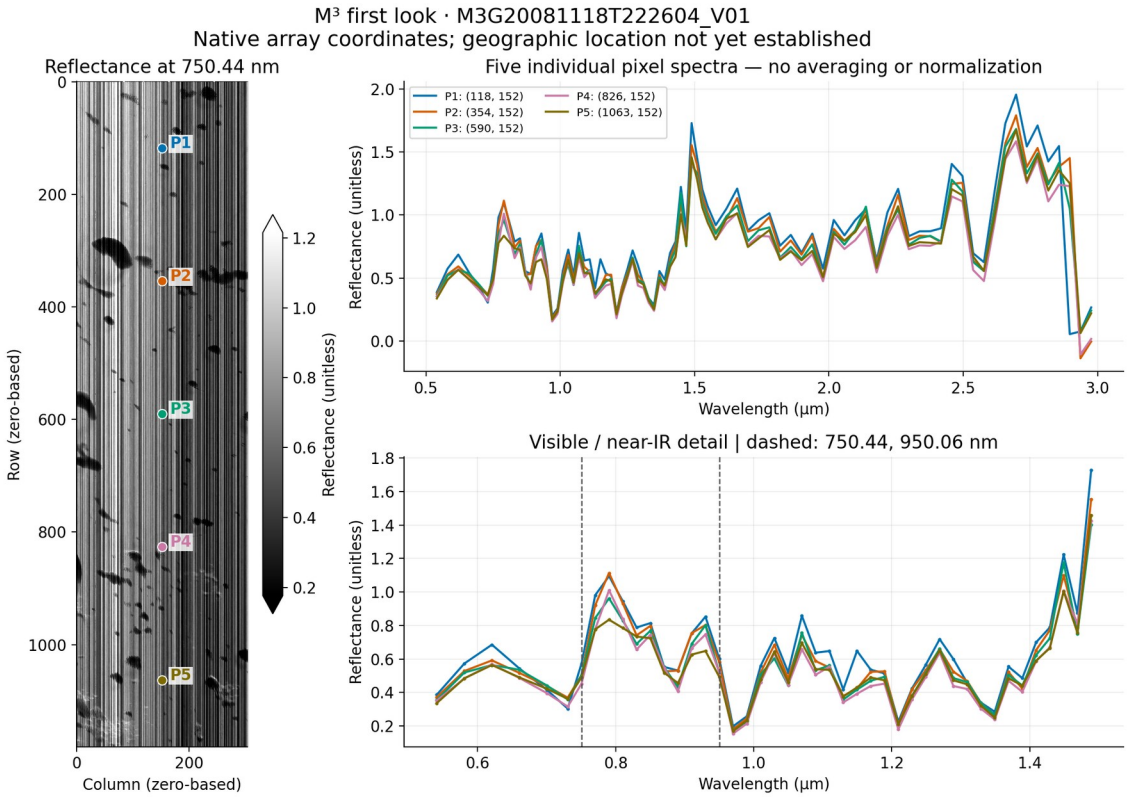


Figure 1. Observation M3G20081118T222604_V01. Left: all image pixels at 750.44 nm. Upper right: five locations measured across wavelength. Lower right: the same spectra from 540 to 1,500 nm; dashed lines locate two potential FeO inputs. These are native image coordinates, not a lunar map. [1, 2]

What has actually been achieved

The script reads one 1,182 × 304 × 85 reflectance cube, checks its layout, extracts spectra and records diagnostics. A fresh rerun for this revision reproduced the entire saved inspection.json exactly, including spectra, statistics and input hashes. This confirms reproducibility with the same code and inputs; it does not independently validate the reader or chemistry. Strong striping and shared spectral irregularities remain unresolved. [1, 2, 8]

The next result should be a verified spectrum and geographic footprint, followed by a tested FeO estimate—not a concentration map made from unvalidated values.

What the graph means and what the code does

One measurement appears in two places

The cube can be understood as 85 wavelength images of the same scene. At a fixed image location, those 85 values form a spectrum. P1 is row 118, column 152. At 750.44 nm its value is 0.5722: this sets its gray shade in the left image and the height of its blue curve at that wavelength. At 950.06 nm its value is 0.6028. Moving right on the graph changes wavelength, not surface position. Moving across the image changes position, not wavelength. [1, 2]

Higher on the curve means a larger stored radiance factor under the product's reflectance convention. It does not mean a higher iron percentage. Dips may be candidates for absorption features, but they are not automatically mineral identifications. The connecting lines join measured channels; they are not a chemical fit. [1, 4, 5, 6]

The operations that produce the figure

Code in inspect_cube.py	What it accomplishes
Lines 41–57 Metadata and assertions	Read RFL.HDR and L2.LBL. Check dimensions, BIL layout, 32-bit storage, header byte order, missing value, wavelength order and expected byte count.
Lines 59–60 memmap and transpose	Read little-endian float32 values in stored order (line, band, sample); expose them as cube[row, column, band]. This rearranges axes, not measured values.
Lines 62–79 Checks and masks	Compare five direct binary reads with array entries; exclude fill/nonfinite values and bad bands; count missing, negative and above-one values.
Lines 80–90 Image and spectra	Find channels nearest 750/950 nm. cube[:, :, b750] selects the image; cube[row, col, :] selects a spectrum. Five evenly spaced interior rows use column 152.
Lines 93–117 Plotting	Draw the image and colored locations, all five spectra, and a 540–1,500 nm detail. Save first_spectra.png.
Lines 119–129 Run record	Save wavelengths, numerical spectra, statistics, byte counts and SHA-256 fingerprints to inspection.json.

The grayscale display stretches the 2nd–98th percentile range, here 0.1768–1.2146. This changes visual contrast, not spectral values. The first two bands are missing and flagged bad. Other negative and above-one values remain visible for diagnosis. Neither smoothing nor spectral normalization is performed. [1, 2]

What these checks do not prove

The five direct reads share the script's layout assumptions; they are not a second reader. The VRT wavelength difference is recorded (0.005 nm maximum), but no agreement tolerance is enforced. The code assumes zero data offset. Some generated report sentences, including correction status and location-file absence, are fixed text rather than tested conditions. The source label and directory were separately inspected for this revision; future code should test these conditions explicitly. [1–3, 8]

How these spectra could establish iron abundance

The physical connection

Iron in certain mineral structures absorbs light over characteristic wavelength ranges. The reading plan describes Fe^{2+}-related absorption near 1 µm and how mineral environment influences spectral shape. Reliable absorption patterns can therefore support an interpretation of Fe-bearing minerals. But mixtures, grain properties, illumination and space weathering also affect brightness and absorption. A dark pixel or a single dip cannot uniquely specify composition. A weak feature also does not establish an absence of iron. [4, physical basis; 5]

First decide whether the measurements are interpretable

The present image has strong column-like stripes, and the five curves share pronounced jagged structure. The cause is unknown. At 750.44 nm, the rerun counts 149 negative values and 25,348 values above one among 359,328 pixels. At 950.06 nm the corresponding counts are 139 and 110,508. These are diagnostic counts, not automatic rejection counts: radiance factor is not a total reflected-energy fraction. [2, 6, 8]

Next, compare numerical values and wavelengths with an independently configured compatible reader. Inspect spectra across multiple columns, quality flags, SUP/geometry data and a second observation. If the same artifacts survive independent reading, investigate the product and acquisition rather than adjusting the plot until it looks smooth. Only a justified usable subset should enter a chemical estimator. [4, 5; proposed experiment]

Then reproduce a specified FeO calibration

The initial literature target uses both brightness R750 and the contrast ratio R950/R750. For P1, that ratio is 1.05345. This is an optical ratio, not 105% iron. The proposed calculation translates the point in brightness–ratio space by an empirical origin, calculates an angle, then converts that angle to FeO-equivalent weight percent. [2, 4, 5]

$x = R750 - 0.08$; $y = R950/R750 - 1.19$
$\theta = -\operatorname{atan2}(y, x)$, in radians
Estimated FeO wt% $= 17.427\,\theta - 7.565$

This signed-angle expression is an implementation candidate to audit, not a resolved reproduction of the 2026 study. That abstract gives an ambiguous wrapping/sign convention and reports a revised origin and refit without a complete numerical recipe. The exact scene, processing, angle code, revised coefficients and sample matching must be obtained. Do not insert guessed coefficients or clip unrealistic predictions into a plausible range. [4, 5, 7]

Finally test whether it predicts known chemistry

Match independently measured soil chemistry to appropriate image footprints. Reproduce the documented baseline before refitting. Evaluate withheld stations or regions, report bias/MAE/RMSE in FeO percentage points, and test sensitivity to channel choice, corrections and registration. Calibration stations reused for comparison provide consistency evidence, not independent predictive validation. [4]

A defensible outcome is “estimated FeO-equivalent wt% under this calibration, with these errors and exclusions.” It is not “this pixel is pure iron,” a mineral fraction or an extractable metal yield. A binary iron-rich classification would need an explicit, scientifically justified threshold after validation. [4, 5; proposed reporting rule]

Why coordinates are missing and how to make a map

The reflectance array does not contain the complete location lookup

The script reads reflectance and plots row/column indices with imshow. It never loads per-pixel longitude/latitude or performs a geographic transformation. M³ has an associated location product; the local download is incomplete for mapping. Renaming the axes would not supply the missing information. [1, 3]

The actual Level 2 label points to these Level 1B version 3 companions: [3]
M3G20081118T222604_V03_LOC.IMG
M3G20081118T222604_V03_OBS.IMG

The LOC product supplies pixel location information—longitude, latitude and radius—while OBS supplies observing geometry. Neither is present in the inspected cube set directory. The V01 reflectance product explicitly references V03 companions; the differing version suffix is not itself an error. Their headers/labels are needed to establish storage, band order, units and invalid values. [3, 8]

The concrete mapping procedure

1. Find the exact LOC and OBS filenames in the official PDS Level 1B archive/index. Record their URLs, versions and acquisition association. Download their metadata and data with byte counts and integrity receipts. This acquisition is pending.

2. Read the location product according to its own metadata, then verify its line/sample alignment with the 1,182 × 304 reflectance grid. At a valid (row, column), pair the spectrum and later FeO estimate with the matching longitude and latitude.

3. Plot the footprint on lunar context imagery and check orientation and recognizable landmarks. Confirm coordinate conventions. The archive specifies planetocentric latitude, the Mean Earth/Polar Axis frame and east-positive longitude from 0 to 360 degrees. Do not assume an Earth coordinate system. [6]

4. Make a located reflectance preview before chemistry. This tests placement separately from estimation. It also determines whether this observation covers Apollo 17; that coverage remains unestablished.

5. After calibration validation, attach coordinates to each accepted FeO value. Choose a lunar projection, output spacing and resampling method appropriate to the footprint. Verify exported values and orientation against native pixels; a finer output grid does not create new spatial information.

6. Export the FeO estimate with no-data/quality flags, coverage and available uncertainty or sensitivity layers. Keep excluded pixels missing rather than zero. Preserve product ID, calibration version and resampling settings.

The first map target is one verified swath. Extending to a larger area then requires additional observations, compatible processing and explicit overlap/seam rules. A coordinate lookup solves placement; it does not solve calibration or data quality. [4; proposed workflow]

Files next steps and the limits of the claim

What each downloaded file contributes

RFL.IMG stores the reflectance measurements; RFL.HDR describes array layout, wavelengths and bad bands; L2.LBL identifies the observation, processing and companions. VRT files describe raster access. SUP contains three diagnostic images, now confirmed from the actual label: 1489 nm reflectance corrected relative to a sphere, estimated surface temperature in kelvin, and Level 1B radiance band 84. They can assist investigation of illumination/thermal behavior, but they are not Fe/Al maps or a substitute for LOC. The script hashes SUP files but does not analyze their pixels. [1, 3]

The label records thermal correction enabled, a photometric correction file and no ground-truth correction file. This establishes processing provenance, not the accuracy of those corrections for this observation. Local hashes record input identity; comparison against publisher checksums remains separate. [2, 3]

Immediate deliverables

Next experiment	Deliverable and decision
Independent reading and quality	Fixed-pixel numerical comparison, per-band/column diagnostics and second-scene comparison. Determine which measurements are suitable for inference.
Recover location and geometry	Verified companion inventory and footprint with coordinate conventions. Decide whether this scene is relevant to the intended study area.
Audit the FeO method	Confirmed calibration specification, synthetic angle checks and reference soil table. Revised 2026 fit stays unresolved until its parameters are known.
Validate then map	Held-out errors and sensitivity results; one georeferenced estimate layer with masks. Broader mapping follows only after this works.

Aluminium and automation

Al requires a separate calibration and validation study. Fischer's relationship in the equation register uses 0.76 µm reflectance relative to MS2 at a specified phase, not a raw M^3 channel. Its historical error cannot be assigned to our map. Do not infer Al by relabeling FeO, assuming a universal Fe–Al anticorrelation or assigning chemistry to a cluster. [5]

Build the deterministic reader, quality checks, calibration, validation and mapping first. Agents can later repeat discovery, downloads and reporting with recorded inputs and settings. ML should test whether extra spectral information improves held-out predictions beyond the baseline; neighboring pixels sharing one coarse reference label are not independent training samples. No such automation or model is completed here. [4, 5]

Evidence used and materials for the meeting

[1] inspect_cube.py, lines 41–129: reviewed implementation. [2] results/first_look/inspection.json and first_spectra.png: numerical/visual baseline. [3] actual M3G20081118T222604_V01_L2.LBL and SUP.HDR: product roles and companion references. [4] M3_first_four_weeks_research_plan.md: physics, replication and validation protocol. [5] equation_register.md and research_system_design.md: calibration limits and separate Al/automation branches. [8] revision check: unchanged script rerun against the local cube set; all JSON content matched the existing baseline. No second-reader or chemistry validation was performed.

[6] NASA PDS Level 2 dataset description, Processing and Coordinate System: https://pds.nasa.gov/ds-view/pds/viewProfile.jsp?dsid=CH1-ORB-L-M3-4-L2-REFLECTANCE-V1.0
[7] Trevino et al., 2026 LPSC abstract 1102: https://www.hou.usra.edu/meetings/lpsc2026/pdf/1102.pdf

New executable result

The rewritten prototype was run on 359,328 native pixels. 359,138 were calculable and 190 were excluded for nonpositive inputs. Using the original published constants literally, the signed branch produced 288,637 negative outputs (80.4%), while a wrapped branch produced 16,979 outputs above 100 wt%. This is evidence that the

calculation runs and that calibration transfer remains unresolved; it is not a validated iron result. The original and wrapped branches are both retained rather than selecting a visually attractive answer.

The script, configuration, machine-readable arrays, five diagnostic figures, selected pixel CSV, run manifest and 11-test record are published in outputs/m3-feo-prototype. The exact command is: python3 src/inspect_cube.py --data-dir "/Users/max/Desktop/M3 UGR/cube set" --output-dir results/new_run. The raw cube remains outside the package.

The next concrete pipeline is: product discovery and receipt → immutable download manifest → independent reader comparison → LOC/OBS alignment and lunar coordinates → FeO calibration reproduction against reference samples → masked georeferenced swath → separate Al calibration branch → only then ML and agent automation. The proposed operator is Codex now and one OpenAI Agents SDK Python runner later; no agent service is deployed.